



THE SMART MOVE

Grade 09

ALIGNED WITH INTERNATIONAL STANDARDS

This book supports essential educational frameworks that shape future-ready minds:

- 21st Century Skills
- Sustainable Development Goals (SDGs)
- Social-Emotional Learning (SEL)
- Universal Design for Learning (UDL)

STEAM Scouts is a ground-breaking STEAM book series with an interdisciplinary focus designed to ignite curiosity, creativity, and critical thinking in learners worldwide. The series supports Social-Emotional Learning (SEL) by promoting core competencies like self-awareness, self-management, social awareness, relationship skills, and responsible decision-making. It also aligns with the UN Sustainable Development Goals (SDGs), fosters 21st Century Skills, and meets international educational standards to equip learners with tools for a dynamic, innovative, and impact-driven future. Furthermore, it is built on Universal Design for Learning (UDL) principles, ensuring inclusive, accessible, and engaging STEAM education for all students, regardless of learning ability.

Key Features

Standards-Aligned

- NGSS (Next Generation Science Standards) – Authentic scientific inquiry & engineering design.
- CCSS (Common Core State Standards) – Rigorous literacy & math integration.
- Cambridge & IB (PYP/MYP/DP) – Global perspectives & interdisciplinary learning.
- ISTE (Technology Standards) – Digital literacy & computational thinking.
- STEL (Standards for Technological & Engineering Literacy) – Real-world problem-solving.

Social-Emotional Learning (SEL) and Global Citizenship

- Each book fosters collaboration, resilience, leadership, and ethical decision-making while connecting to multiple UN Sustainable Development Goals (SDGs) - such as Quality Education (SDG 4), Clean Energy (SDG 7), Climate Action (SDG 13), and Innovation (SDG 9) - empowering students to think globally, encourages ethical innovation & community impact.
- Differentiated challenges to meet diverse learning needs and skill levels.
- Visual cues and infographics to enhance understanding and engagement.
- Guided instruction to support learners through new or complex concepts.
- Scaffolded activities that build skills progressively and promote learner independence.

Why Choose This Series?

Future-Ready Skills – Prepares learners with essential skills like creativity, collaboration, and critical thinking for future careers and global challenges.

Teacher & Student-Friendly – Includes engaging lesson hooks, discussion prompts, guided instruction, and hands-on tasks for seamless learning.

- **SDG-Themed Modules** – Each module links to 2+ UN SDGs, blending academics with global purpose and ethical action.
- **Inclusive & Engaging** – Built on UDL principles to ensure accessibility through differentiated pathways, visual supports, and hands-on learning.

Perfect For:

- K-12 Schools adopting IB, Cambridge, NGSS, or CCSS
- STEM/STEAM Coordinators seeking industry-relevant curricula
- Homeschoolers and Educators wanting SDG-aligned, future-focused content
- Educational Institutions committed to sustainability and inclusion

Theme

03

THE SMART MOVE

Each week has a double session, total time of session 80-90 mins



Week	Unit Name	Lesson	Page
Week 1	Unit 1: Spark The Challenge!	Lesson 1: Beyond Barriers: Tech for Independent Living.	2-15
Week 2		Lesson 2: Design for Independence.	16-27
Week 3		Lesson 3: Rolling Forward.	28-44
Week 4	Unit 2: Innovators in Action.	Lesson 1: Scaling up for Reality.	45-59
Week 5		Lesson 2: The Future of Mobility.	60-83
Week 6	Unit 3: Beyond Imagination.	Lesson1: Seamless Screens.	84-99
Week 7		Lesson 2: Command and Move.	100-123
Week 8		Lesson 3: Prototype to Pitch.	124-139

About The Theme

Everyone deserves the freedom to move, explore, and live independently—regardless of physical ability. Yet for many individuals with mobility challenges, navigating daily life can be difficult, especially in environments that lack accessible design. But what if technology could do more than assist—what if it could empower? In this theme, students will become real-world innovators as they design and build a Smart Wheelchair that provides not just movement, but autonomy, safety, and intelligent control.

The journey begins with students creating a small-scale wheelchair prototype that uses joystick control, then evolves into a fully functional assistive device integrated with Bluetooth and mobile app control. But the real leap forward comes with the use of artificial intelligence: students will add voice command functionality, allowing users to control the wheelchair through spoken instructions—a meaningful application of AI-driven speech recognition that brings independence to a whole new level.

As they build, students will explore the physics of motion, including how force, mass, torque, and balance impact mobility. They'll program motors using Arduino, connect electronic components, and refine mechanical designs to improve stability and responsiveness.

Students will also develop custom apps using MIT App Inventor, enabling wireless control, real-time feedback, and safety monitoring. With a strong focus on user experience, they'll design features that prioritize accessibility, comfort, and ease of use—considering the real needs of people with limited mobility.

Through this engaging, multidisciplinary challenge, students will develop essential skills in:

- Arduino programming and motor control.
- App development and Bluetooth communication.
- AI integration through voice recognition and speech-to-text.
- Empathy-driven, human-centered engineering.
- Iterative design, testing, and problem-solving.

Aligned with Sustainable Development Goals 3 (Good Health and Well-being) and 10 (Reduced Inequalities), this project empowers students to build not just a robotic device—but a smart, inclusive tool that enhances freedom and dignity. By the end of the theme, students will have created a prototype that demonstrates how AI, coding, and compassionate design can improve lives and redefine what's possible for everyone.

HOW TO USE THIS BOOK?

1

Ready, Set, Learn!

Where we merge your existing knowledge with exciting new discoveries.

2

Inquiry-Based Learning!

Let's Think!

In this section, students will brainstorm and explore possible solutions to specific problems. This encourages critical thinking and creativity as they consider various approaches.

Explorer's Toolkit

Here, students will dive deeper into the topic. They will engage with curated videos, articles, educational games, and hands-on experiments to broaden their understanding and spark curiosity.

DIY Lab

It's part of the explore, students will undertake hands-on projects where they will design, build, and test their own creations. They will apply scientific principles, use technology and engineering skills, explore artistic concepts, and utilize mathematical reasoning.

Screen Lessons

Students will watch videos that explain key concepts in an engaging and accessible way.

Dig Deeper

Students will read articles that provide deeper insights into specific concepts, enhancing their understanding.

3

Problem-Based Learning:

- **Introducing the problem:**
Students will join a journey to tackle real-world environmental challenges. Through hands-on activities and engaging tasks, they will apply their knowledge and skills to discover innovative solutions.
- **Apply the Engineering Design Process (EDP):**
Students will use a structured approach to define problems, conduct research, generate ideas, prototype solutions, test, evaluate, and refine their designs.
- **The Scientific Method:**
Students will follow a systematic process of inquiry—making observations, forming hypotheses, conducting experiments, analyzing data, and drawing conclusions—to develop a deeper understanding of scientific concepts and problem-solving. In doing so, they will incorporate the CER framework to present their scientific claims, provide evidence from their experiments, and explain their reasoning behind the conclusions they draw.

B

4

Project-Based Learning:

PBL immerses students in real-world challenges through hands-on projects. They build critical thinking, creativity, collaboration, and communication skills while designing solutions and taking ownership of their learning.

Case Study

The journey shifts from storytelling to real-world problem-solving. Through engaging case studies, students will analyze real-life challenges, explore innovative solutions, and apply STEAM principles to tackle complex issues. By investigating real-world scenarios, they will develop critical thinking, problem-solving, and decision-making skills, preparing them for future careers and global challenges.

5

Assessment Exercises:

These are divided into three categories to ensure comprehensive learning and mastery.

Apprentice:

Developing understanding and can apply skills with guidance.

Builder:

Demonstrates solid understanding and can apply skills independently.

Mastermind:

Demonstrates deep understanding and can apply skills creatively and strategically.

6

Career Readiness

This section equips students with essential skills for future careers by fostering problem-solving, teamwork, and adaptability. Through hands-on projects, real-world applications, and critical thinking challenges, students develop the confidence and competencies needed for success in the ever-evolving job market.

7

Mission Innovate

Now, it's your turn! In Mission Innovate; students will apply their creativity, problem-solving skills, and knowledge to develop their own competition projects. This challenge encourages independent thinking, collaboration, and real-world innovation, empowering students to become the next generation of creators and problem-solvers.

Now I Can

This section is designed to help you measure your progress and achievements. Enjoy your exploration and learning journey through this STEAM book!

My Journal

In this part students will write their own story while they are creating the project.

By the end of this theme, students will be able to:

- Understand common mobility-related challenges experienced by people with physical limitations.
- Use empathy tools (like interviews and mapping) to define user-centered problems.
- Apply the Engineering Design Process to develop and improve assistive solutions.
- Calculate force, torque, and efficiency in real-world movement scenarios.
- Choose materials and structural designs using ergonomic and scientific reasoning.
- Integrate Technology: Build and program electronic systems and mobile applications to create functional smart mobility prototypes.
- Analyze Scaling Impacts: Evaluate how design decisions made during prototyping affect the feasibility, cost, materials, and performance when scaling a product for real-world use.
- Communicate Design Solutions: Clearly present their design processes, technical reasoning, and proposed solutions for complex engineering challenges.
- Collaborate in teams to brainstorm, build, test, and iterate design solutions.
- Communicate your ideas clearly through presentations, diagrams, and simple code explanations.
- Reflect on how your design process supported real-world needs and user independence.

Unit

1

Lesson

1

Spark The Challenge!

Beyond Barriers: Tech for Independent Living.

Time Frame: [90 min]

Lesson Objectives:

By the end of this lesson students will be able to:

- Identify real barriers people with mobility needs face
- Practice gathering user data through observation and interviews.
- Create an Empathy Map to organize user insights
- Begin thinking like problem-solvers designing with purpose.

Challenge questions of the lesson

1. Why is empathy crucial when designing technology for people with mobility challenges?
2. Name one specific challenge a person with limited mobility might face in an everyday setting.
3. How can a smart technology (like a sensor or motor) make a mobility solution more effective?
4. Give an example of a "design trade-off" you might consider when creating a new assistive device.

SEL Relation

- **Community Awareness:** Understand how smart technology fits into improving accessibility and independence for people with mobility challenges in the community.
- **Responsibility:** Recognize our personal role in promoting inclusive design and contributing to a more accessible world for everyone.
- **Collaboration:** Work together during design challenges, discussions, and prototyping to foster innovative solutions for accessible futures.
- **Appreciation:** Value smart technologies and the enhanced quality of life achieved through effective inclusive design and assistive technology.
- **Self-Awareness:** Reflect on learning about smart inclusive design and identify areas for personal growth in promoting accessibility and empathy.
- **Communication:** Clearly express ideas about smart mobility solutions through speaking, drawing (sketches/prototypes), and writing (design briefs).
- **Critical Thinking & Problem-Solving:** Analyze mobility challenges, develop solutions through design thinking and prototyping, and creatively address accessibility needs.
- **Design Literacy/Responsibility:** Deepen understanding of how design impacts accessibility and independence and take action through empathy-driven, smart inclusive design.

Engage



Teacher's Role:

- **Facilitator:** Guide a discussion about everyday challenges faced by individuals with mobility challenges.
- **Questioning:** Ask guiding questions to encourage critical thinking about user needs and potential tech solutions.
- **Presenter:** Introduce the problem-solving approach by connecting real-world mobility challenges to potential smart technology solutions.

Expected From Students:

- **Read the Case:** Carefully review the case study, focusing on the individual's daily life and specific mobility difficulties.
- **Reflect on Key Points:** Identify the main obstacles and how they impact the individual's routines or well-being.
- **Ask Clarifying Questions:** Think about what more you'd need to know to fully grasp their experience (e.g., most difficult tasks, frustrations with current aids).
- **Connect to Personal Understanding:** Consider how your own day would change with similar limitations to build empathy for their situation.

Parts of the book included.

Case Study

Project overview:

Imagine a world where daily life flows more smoothly, more conveniently, and more sustainably for everyone. A world where everyday frustrations are transformed into clever solutions, and where technology truly serves people. Now, imagine you are the team responsible for helping create that world.

Your challenge is to step into the shoes of an inventor and design a smart technology solution that tackles an everyday problem head-on, making life easier and better for individuals and communities.

Over the coming weeks, you'll embark on an exciting journey. You'll uncover real-world annoyances, explore the fascinating world of sensors, motors, and displays, and apply your creativity and problem-solving skills to design impactful solutions. Get ready to combine engineering, coding, design thinking, and a healthy dose of empathy to develop tech-based innovations that respond to the genuine needs of real people. Let's make daily life brilliantly smart!

Mission:

You've been invited into a real-world challenge where science, design, and empathy come together.

Your task is to research, design, and prototype a tech solution that supports mobility, safety, and independence for people who face physical challenges in everyday life.

The solution should:

- Adapt to both indoor and outdoor spaces.
- Be simple and intuitive to control.
- Offer safety features that increase confidence and awareness.
- Enhance comfort and dignity for the user.

Your final creation will be featured at the Tech for Good Youth Expo, where teams from across the country will share inclusive technologies designed to improve lives.

Driving Questions:

1. What real-world barriers do people with physical mobility challenges face in their daily environments?
2. How can technology—like sensors, motors, and smart interfaces—be used to improve safety, independence, and comfort for users with limited mobility?
3. What makes a smart mobility solution both effective and easy to use for people of all abilities?
4. How can we design a product that adapts to different spaces (indoor and outdoor) while remaining safe and reliable?
5. How do empathy and user research help inventors build more inclusive, human-centered technology?

STEAM SCOUTS

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Pyramakerz®

6. What engineering and coding skills are needed to prototype a functional smart device that responds to real-world needs?
7. How can we communicate our design choices and impact clearly to inspire others at the Tech for Good Youth Expo?

Let's Think!



What everyday challenges do people with reduced mobility face that others might overlook?

How do physical changes like muscle weakness, slower movement, or limited vision impact independence and safety?

Explorer's Toolkit



DIY Lab

Can you move through the world like someone with a mobility challenge?

You'll work with a partner to navigate an obstacle course. One of you will simulate limited leg movement using tape, while the other stays mobile. Together, you'll find ways to complete the challenge—then reflect on how teamwork helped you succeed.

Goal: Reach the finish line and reflect on what it felt like.

Think About It:

- What made the task hard?

- How did your partner help?

STEAM SCOUTS

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Case Study



1. Read the Case Study Together

- Display and read the Project Overview aloud with energy and clarity.
- Pause at the line: “Now, imagine you are the team responsible for helping create that world.”
- Ask: “What kinds of everyday problems do you think people with mobility challenges face that others may not even notice?”
- Encourage students to reflect on how small inconveniences for some can be big obstacles for others.

2. Introduce the Mission

- Read Your Mission aloud.
- Ask: “Why is it important that solutions are both functional and respectful for users?”
- Have students brainstorm a few real-life situations where someone with mobility challenges might struggle—like navigating public spaces, adjusting furniture, or transferring from a chair to a bed.
- Record student responses on the board or a shared chart.

3. Guide Inquiry with Driving Questions

- Display the Driving Questions on the board or screen.
- Choose 1–2 to discuss as a group. For example:
 - “How can smart technology make movement safer and more independent for people with physical challenges?”
 - “What makes a tech solution truly user-friendly for all abilities?”
- Encourage students to draw from real-life experiences, media, or personal observations.

4. Set the Challenge

- Explain that in this unit, students will:
 - Identify a real-world mobility challenge faced by individuals with spinal cord-related physical limitations.
 - Use empathy mapping and design thinking to understand the user's needs.
 - Design and prototype a smart tech solution that combines sensors, automation, and user-centered design.
- Emphasize: “This isn’t just about building a project—it’s about building solutions that matter to real people. You’re the inventors. You’re the changemakers.”

Possible Strategies for Implementation:

1. User-Centered Interviewing:

- Strategy: When talking to individuals with mobility challenges, focus on open-ended questions. Ask about their daily routines, frustrations, joys, and aspirations. Listen more than you talk.
- Why: Captures authentic experiences, unmet needs, and insights that might not be obvious.

2. Observational Research ("Shadowing"):

- Strategy: Observe how individuals with mobility challenges navigate different environments (home, public spaces, outdoors). Note their workarounds, points of struggle, and moments of ease.
- Why: Reveals real-world interactions and environmental barriers that users might not articulate in an interview.

3. Brainstorming Diverse Solutions:

- Strategy: Encourage wild, unconstrained ideas initially. Don't judge too early. Think beyond obvious tech and consider simple, non-tech, or hybrid solutions first.
- Why: Fosters creativity and leads to truly innovative, out-of-the-box thinking.

Let's Think!



Objective:

Students will develop a deep and empathetic understanding of the everyday challenges faced by individuals with reduced mobility, exploring how physical changes impact their independence and safety, and to identify opportunities for designing smart, simple, and supportive tech solutions.

Answers:

1. What everyday challenges do people with reduced mobility face that others might overlook?

- Difficulty reaching high shelves or low drawers
- Trouble navigating curbs, stairs, or uneven paths
- Limited access to public transportation or restrooms
- Delays crossing streets due to shorter walk times at signals
- Struggling with doors that don't open automatically

2. How do physical changes like muscle weakness, slower movement, or limited vision impact independence and safety?

- Muscle weakness can make it hard to push a wheelchair or transfer to a bed/chair safely
- Slower movement may increase the risk of falls or not reacting quickly in emergencies

- Limited vision can make it difficult to avoid obstacles or read signs, increasing dependence on others
- These changes can make everyday tasks—like cooking, shopping, or navigating unfamiliar areas—more dangerous or exhausting

Possible Strategies for implementation:

Empathy Simulation:

- Strategy: Engage in activities that simulate mobility challenges (e.g., navigating an obstacle course with restricted movement, using a wheelchair, or attempting tasks with limited vision).
- Why: Provides firsthand experience, revealing unexpected difficulties and fostering deeper empathy.
- Strategy: After gathering insights, clearly define specific problems using "How Might We...?" questions. Then, brainstorm a wide range of smart gadget ideas without immediate judgment, focusing on how technology (sensors, motors, displays) can address those identified needs.
- Why: Ensures the design phase is targeted at genuine problems and encourages innovative, tech-driven solutions.

Explorer's Toolkit



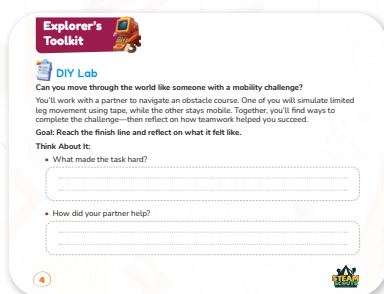
Teacher's Role:

- Prepare a safe, moderately challenging obstacle course using classroom furniture and materials.
- Group students in pairs and assign roles: one student simulates limited mobility, the other offers support.
- Explain safety rules and clarify expectations (e.g., stay within arm's reach, collaborate, finish the course).
- Turn the simulation into a friendly challenge to encourage teamwork.
- Lead a whole-class reflection discussion and distribute empathy maps for deeper insight.

Expected From Students:

- Collaborate respectfully with their partner to navigate the obstacle course.
- Take turns in the mobility-restricted role and support role.
- Stay safe, communicate clearly, and complete the challenge using teamwork.
- Reflect on the experience, both emotionally and practically, through empathy maps or discussion.

Parts of the book included.



DIY Lab

Objective:

Students will develop deep empathy for individuals with mobility challenges by simulating their experience, collaborating to overcome obstacles, and using Empathy Maps to capture insights that will inform innovative design solutions.

Materials Needed:

- Masking tape (to restrict one leg per student or 2 legs)
- Cones, backpacks, chairs, or other classroom items for maze setup
- Stopwatch or timer
- Empathy map templates
- Scenario cards (optional, for deeper discussion)

Activity Instructions:

1. Prepare the Course

- Use classroom materials to create a pathway with mild obstacles (turns, low objects, narrow spaces).

2. Pair Up Students

- In each pair, one student wraps one leg with masking tape to simulate reduced mobility.
- The partner assists and supports—verbally or physically—but cannot complete the course alone.

3. Set the Challenge

- Give each pair 5 minutes to complete the obstacle course.

EMPATHY MAP

SAYS	THINKS
• "Can someone give me a hand?" • "Wait, I can't reach that!" • "This is harder than I thought." • "I feel stuck here." • "I don't want to slow us down." • "How do people do this every day?" • "Let's try it another way."	• "I didn't realize how many obstacles there are." • "This would be so frustrating in real life." • "I wish things were easier to access." • "Even small tasks feel big right now." • "My teammate is being really patient—thankfully!" • "Design should make life easier, not harder."
DOES • Holds onto furniture for support. • Asks partner to grab an item. • Crawls or scoots instead of walking. • Pauses often to figure out next steps. • Tries different angles to reach things. • Communicates actively with the teammate. • Avoids certain obstacles entirely.	FEELS • Frustrated at not being able to move freely. • Grateful for their teammate's help. • Embarrassed about struggling. • Determined to finish the challenge. • Surprised by how hard it actually was. • Curious about how real wheelchair users feel. • Inspired to create better designs for accessibility.



- Optionally, make it a competition: the fastest safe team wins.

4. Switch Roles (Optional)

- If time allows, let students swap roles for a second round.

5. Reflect

- After completing the task, provide students with empathy maps and invite them to fill them out.
- Facilitate a whole-class or small group reflection using the provided questions.

Possible Strategies for Implementation:

1. Focused Observation & Self-Reflection:

- How to implement:
During the obstacle course, the mobile partner should meticulously observe the partner simulating limited mobility. Both students should note specific difficulties, frustrations, and unexpected insights. This direct observation and immediate self-reflection are crucial for gathering raw data for the Empathy Map.

2. Guided Empathy Mapping:

- How to implement:
After the challenge, use questions like "What made the task hard?" and "How did your partner help?" as a direct springboard for completing the Empathy Map. Encourage partners to help each other fill out the "Says," "Thinks," "Does," and "Feels" sections based on their shared simulated experience.

3. Connecting to Design:

- How to implement:
As part of the reflection and Empathy Map discussion, explicitly ask: "How does this experience and what we learned from our Empathy Map inform how we might design a smart solution for this challenge?" This bridges the empathy-building activity directly to the next phase of the engineering design process, guiding students towards user-centered problem-solving.

Explain



Teacher's Role:

- **Facilitator:** Guide students to focus their attention on key aspects of the videos (e.g., specific challenges, emotional impact, existing solutions shown).
- **Questioning:** Ask probing questions during and after video segments to prompt critical thinking and initial empathy, such as "What did you observe?", "How might this impact daily life?", or "What problems did you notice?"
- **Presenter:** Set the context for each video, highlighting its relevance to understanding mobility challenges and inspiring thinking about potential smart solutions.

Expected From Students:

- Actively watch the provided video links with a purpose (e.g., to identify specific challenges, observe coping mechanisms, or spot existing technologies).
- Take notes on specific difficulties, unmet needs, or ingenious workarounds observed in the videos.
- Share initial reflections on how these video insights contribute to their understanding of mobility challenges.
- Connect observations from the videos to the broader goal of designing tech solutions.

Parts of the book included.



Screen Lessons

Let's learn more about the needs of people with mobility challenges:

<https://www.youtube.com/watch?v=Ag9-9TF8Wel>

<https://www.youtube.com/watch?v=Dy7KlgzWNUA>



Screen Lessons

Objective:

Students will identify and understand diverse real-world challenges faced by individuals with mobility limitations by critically observing video case studies, fostering initial empathy, and recognizing the spectrum of needs that technology can address.

Video link: <https://www.youtube.com/watch?v=Ag9-9TF8Wel>

Video link: <https://www.youtube.com/watch?v=Dy7KlgzWNUA>



Possible Strategies for implementation:

1. Pre-Watching: Setting a Purpose

- Strategy: Before watching the videos, introduce the topic of mobility challenges and their impact on daily life. Explain that the videos will provide different perspectives and insights.
- How to Implement: Briefly discuss the types of challenges people with mobility limitations face. Ask students to consider what they already know and what they hope to learn from the videos.

2. During Watching: Active Engagement

- Strategy: Encourage students to actively engage with the videos. This could involve note-taking, using a graphic organizer, or focusing on specific aspects (e.g., emotional impact, technological solutions).

3. Post-Watching: Discussion and Reflection

- Strategy: Facilitate a class discussion about the videos. Encourage students to share their observations, insights, and reflections. Connect the video content to the "Team Up, Move Forward Mobility Challenge" and the design project.

- How to Implement: Use open-ended questions to guide the discussion, such as:
 - What were the main challenges faced by individuals in the videos?
 - How did the videos make you feel?
 - What existing technologies or solutions were shown?
 - How could technology improve the lives of people with mobility challenges?
 - How did the videos connect with the "Team Up, Move Forward Mobility Challenge"?

Elaborate and Evaluate



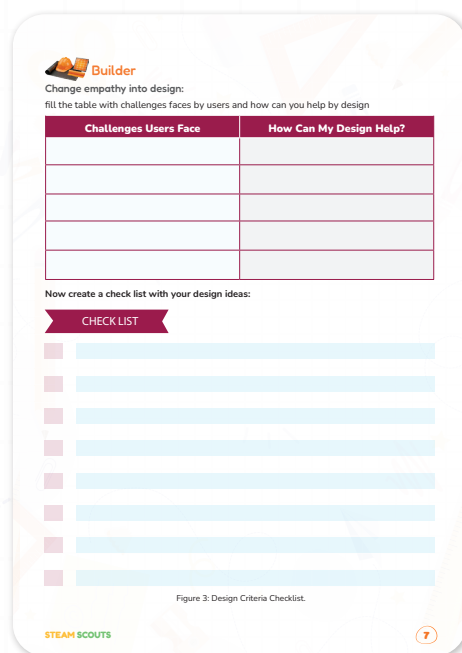
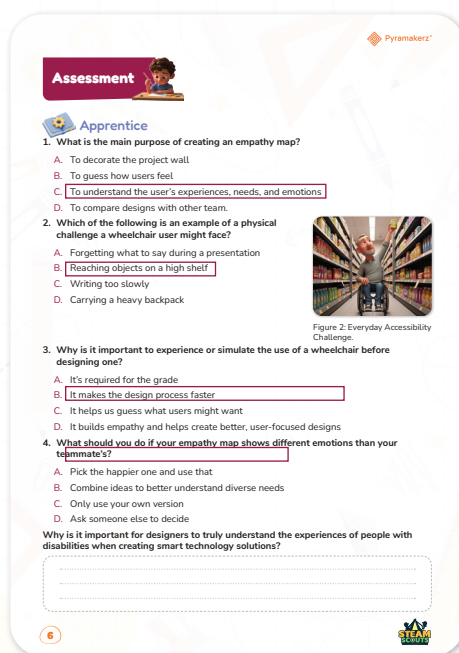
Teacher's Role:

- Guide students through the questions, offering clarification if needed.
- Facilitate a brief discussion after students have completed the questions to review answers and reinforce key concepts.
- Assess student understanding of empathy, mobility challenges, and the design process.

Expected From Students:

- independently answer the multiple-choice questions.
- Demonstrate understanding of empathy maps, the importance of user experience simulation, and collaborative problem-solving.
- Be prepared to explain their reasoning for chosen answers.

Parts of the book included.





Apprentice

Objective:

Students will be able to demonstrate comprehension of:

- The main purpose of empathy maps.
- Common physical mobility challenges.
- The importance of user experience simulation in design.
- Strategies for collaborating with diverse perspectives during empathy mapping.

Answers:

1. What is the main purpose of creating an empathy map?

Correct answer: C. To understand the user's experiences, needs, and emotions

2. Which of the following is an example of a physical challenge a wheelchair user might face?

Correct answer: B. Reaching objects on a high shelf

3. Why is it important to experience or simulate the use of a wheelchair before designing one?

Correct answer: D. It builds empathy and helps create better, user-focused designs

4. What should you do if your empathy map shows different emotions than your teammate's?

Correct answer: B. Combine ideas to better understand diverse needs

Possible Strategies for Implementation:

- Individual Assessment: Students complete the questions independently as a quick check for understanding.
- Think-Pair-Share: Students answer individually, then discuss their answers with a partner before a whole-class review.
- Interactive Quiz: Use an online quiz tool (if available) for immediate feedback, followed by a discussion of commonly missed questions.



Builder

Objective:

Students will apply their understanding of user empathy and mobility challenges to brainstorm, define, and sketch initial smart technology design solutions, translating observed needs into practical concepts.

Materials Needed:

- Pencils, pens, and colored pencils

Activity Instructions:

- Reflect and Connect: Review your Empathy Maps and notes from the "Mobility

Challenge" and video observations. Think about the specific difficulties, frustrations, and unmet needs of people with mobility challenges.

- Fill the Table: Challenges to Design! In the table below, list at least three specific "Challenges Users Face" that you identified. Then, for each challenge, brainstorm and describe "How Can My Design Help?" by outlining a smart technology solution. Think about how sensors, motors, or displays could be used.
- Sketch Your Smart Design Idea: Choose one of your "How Can My Design Help?" ideas from the table above. On a separate piece of paper, create a rough sketch of your smart design solution. Label its key parts (like sensors, motors, display, control interface) and show how it would help the user.
- Create a Checklist for Your Design: Now, create a short checklist specific to your sketched design idea. This checklist should highlight the essential features and considerations for your smart solution.
- My Smart Design Idea Checklist:
 - ☐ What specific problem does my design solve?
 - ☐ What smart features (sensors, motors, display) does it include?
 - ☐ How would the user control it? Is it simple and intuitive?
 - ☐ Does it prioritize user safety? (e.g., provide warnings, ensure stability)
 - ☐ Does it enhance the user's independence or comfort?
 - ☐ Is it adaptable for different spaces (indoor/outdoor)?



Now I Can

Objective:

Students will reflect on and articulate their mastery of key concepts related to understanding mobility challenges, effective assistive technology design, and the application of smart solutions for accessibility.

Activity Instructions:

Step 1: Revisit the Learning Goals

- Activity: Look back at the "Now I can" statements below. These represent the key skills and knowledge you've gained in this lesson about designing for mobility.

Step 2: Student Self-Reflection

- Activity: Individually, read each "Now I can" statement. Consider how well you understand and can explain each point based on what you've learned and done in the lesson. Check the box next to each statement you feel confident in.

Step 3: Small Group Discussion

- Activity: In your small groups, discuss each "Now I can" statement. Share specific examples from the "Mobility Challenge," the videos, or your design activities that illustrate your understanding. Help each other clarify any points that are still unclear.

or provide further examples.

Relation to Standards

1. Next Generation Science Standards (NGSS)

- **Engineering Design (ETS1):** This lesson is heavily rooted in the Engineering Design Process.
 - **HS-ETS1-1:** Analyze a major global challenge to specify qualitative and quantitative criteria and constraints for solutions that account for societal needs and wants. (Applies to defining mobility challenges and user needs as the "global challenge.")
 - **HS-ETS1-2:** Design a solution to a complex real-world problem by breaking it down into smaller, more manageable problems that can be solved through engineering. (Applies to designing smart tech solutions for specific aspects of mobility.)
 - **HS-ETS1-3:** Evaluate a solution to a complex real-world problem based on prioritized criteria and trade-offs that account for a range of constraints, including cost, safety, reliability, and aesthetics, as well as possible social, cultural, and environmental impacts. (Applies to discussing design trade-offs and evaluating prototypes.)

2. ISTE Standards for Students (Technology)

- **1.4 Innovative Designer:** Students use a variety of technologies within a design process to identify and solve problems by creating new, useful or imaginative solutions.
 - **1.4a:** Students know and use a deliberate design process for generating ideas, testing theories, creating innovative artifacts or solving authentic problems. (Directly applies to applying the EDP for mobility solutions.)
 - **1.4b:** Students select and use digital tools to plan and manage a design process that considers design constraints and calculated risks. (Applies if digital sketching/ design tools are used, and to thinking about constraints.)
 - **1.4c:** Students develop, test and refine prototypes as part of a cyclical design process. (Applies to sketching and future prototyping stages for their smart solutions.)
 - **1.6 Creative Communicator:** Students communicate clearly and express themselves creatively for a variety of purposes using the platforms, tools, styles, formats and digital media appropriate to their goals.
 - **1.6a:** Students choose the appropriate platforms and tools for meeting the desired objectives of their creation or communication. (Applies to choosing how to sketch/ present their designs.)
 - **1.6d:** Students publish or present their original works or responsibly repurpose digital resources to create new works. (Applies to presenting their final creations at

the "Tech for Good Youth Expo.")

- **1.7 Global Collaborator:** Students use digital tools to broaden their perspectives and enhance their learning by collaborating with others and working effectively in teams across a global and digital landscape.
- **1.7b:** Students use collaborative technologies to work with others, including peers, experts or community members, to examine issues and problems from multiple viewpoints. (Applies to partner work in the mobility challenge and collaborative design.)